OptiCyte

A Low-Cost Optical Platform for Biological Particle Detection — Technical Methodology & Design Rationale

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Document purpose: This document explains the engineering methodology, design decisions, and validation plan behind OptiCyte, for evaluation by ideathon judges. It distinguishes clearly between what is proposed to be built as a working prototype and what is a longer-term research direction, in the interest of technical transparency.

Estimated bill-of-materials cost (core prototype): approximately Rs. 3,000 (estimated, includes margin for shipping, spares, and part-price variability)

Reference laboratory equivalent: benchtop flow cytometer, typically $50,000 (~Rs. 42,00,000) and above

1. Executive Summary

OptiCyte proposes a low-cost optical sensing platform that distinguishes biological particles (mold spores, pollen, yeast, bacteria) from non-biological particles (mineral dust, debris) using laser-induced light scattering and fluorescence, the same underlying physical principle used in laboratory flow cytometers. The prototype is built from commodity, hobbyist-accessible components — a repurposed 405 nm laser diode, a silicon PIN photodiode, a custom transimpedance amplifier and analog peak-detector circuit, a 16-bit ADC, and an ESP32 microcontroller — targeting a total bill-of-materials cost of approximately Rs. 3,000 (estimated), versus $50,000 and above for comparable lab instruments.

The near-term, buildable goal is real-time biological-vs-mineral particle classification and counting, validated first against agricultural particulates (baker's yeast as a non-pathogenic biological standard, versus mineral dust). This document explains the optical and electronic design in detail, the reasoning behind each component choice, the planned validation methodology, and — separately and explicitly labeled as a longer-term research direction — how the same optical architecture could plausibly be extended toward rapid phenotypic antibiotic-susceptibility screening.

2. Problem Statement

Existing low-cost particulate sensors (e.g., laser-scatter dust sensors used in consumer air-quality monitors) measure *how much* particulate matter is present, but cannot determine *what kind* — they cannot distinguish a biological particle, such as a mold spore, from an inert mineral dust particle of similar size. Instruments capable of this distinction (flow cytometers) exist, but are laboratory-bound, require trained operators, and cost tens of thousands of dollars, putting them entirely out of reach of individual farmers, small greenhouse operators, and school science programs.

This creates a genuine, well-documented gap in agricultural settings: fungal and mold outbreaks are frequently detected only after visible crop damage has occurred, by which point yield loss has already begun. An affordable, field-deployable early-warning sensor could shift this from reactive to proactive management.

3. Core Physical Principle

When a particle passes through a focused laser beam, it scatters light in all directions (elastic scattering, same wavelength as the source). This occurs for both biological and mineral particles and is primarily informative about particle size and shape.

Biological particles additionally **fluoresce** under short-wavelength (violet/UV) excitation: endogenous biomolecules such as NADH, riboflavin, and various structural proteins absorb 405 nm light and re-emit it at longer wavelengths (typically 420–600+ nm). Mineral dust does not fluoresce in this way. By simultaneously measuring (a) elastically scattered light at 90° to the beam and (b) the same channel's response after blocking the excitation wavelength with an optical long-pass filter, the system can distinguish a fluorescence signature (biological) from pure scatter (non-biological, or biological with weak native fluorescence, a known and disclosed limitation — see Section 9).

4. System Architecture Overview

The signal path consists of four functional stages:

- **Excitation:** a 405 nm violet laser diode, switched on/off by the microcontroller, illuminates particles as they pass through a sampling channel.

- **Optical detection:** a silicon PIN photodiode positioned at 90° to the beam, fitted with a 435 nm long-pass optical filter, captures scattered/fluorescent light flashes as particles transit the beam.
- **Signal conditioning:** a transimpedance amplifier (TIA) converts the photodiode's nanoamp-scale photocurrent into a measurable voltage; an analog peak-detector circuit captures and holds the peak voltage of each transient flash, since these flashes are far faster than the ADC's sampling rate can capture directly.
- **Digitization & control:** a 16-bit ADC (ADS1115) reads the held peak voltage over I2C; an ESP32 microcontroller sequences the laser trigger, ADC reads, and peak-detector reset, and reports particle event data.

4.1 Why this architecture, and not a simpler one

A simpler design — reading the photodiode signal directly at a fixed sample rate — was considered and rejected. The ADS1115's maximum sample rate (860 samples/second, ≈1.16 ms per sample) is far slower than the microsecond-to-low-millisecond duration of a single particle-transit light pulse. Direct sampling would alias or entirely miss most events. The peak-detector stage decouples pulse-capture speed from ADC speed: a diode-and-capacitor circuit captures and holds each pulse's peak value long enough (several milliseconds) for the comparatively slow ADC to read it accurately, after which the microcontroller discharges the hold capacitor via a MOSFET switch and the system is ready for the next event. This design decision — identified through analysis of the ADC/optical-pulse timing mismatch rather than by starting with a peak-detector as a default choice — is a key piece of the system's engineering rationale.

5. Component-Level Design Rationale

5.1 Excitation source

A 405 nm, 5 mW (Class 3R) violet laser diode is used, sourced by repurposing the diode and driver circuit from a commercially available multi-color laser pointer pen, rather than purchasing a standalone industrial laser module. This is a deliberate cost decision: a focusable industrial 405 nm module can cost upwards of Rs. 2,000, while a pointer pen containing an equivalent diode (with its own manufacturer-tuned current-regulation driver circuit already built in) is available for approximately Rs. 280–420. The diode is switched via a 2N2222 NPN transistor under microcontroller GPIO control, with a pull-down resistor on the transistor base to ensure the laser defaults to an off state during microcontroller boot/reset, before firmware has taken control of the pin — an explicit safety consideration given this is an optical source.

5.2 Photodetection

A BPW34 (or equivalent, e.g. OSRAM SFH 203P) silicon PIN photodiode is operated in reverse-biased photoconductive mode, positioned at 90° to the laser path. A 435 nm long-pass optical filter sits in front of the photodiode to reject the 405 nm excitation wavelength while passing longer-wavelength fluorescence emission, which is what allows the system to differentiate a fluorescence response from simple scatter.

5.3 Transimpedance amplifier (TIA)

Photocurrent from the photodiode (nanoamp scale) is converted to a measurable voltage by an op-amp configured as a transimpedance amplifier ($V_{out} = -i_{in} \times R_f$). The feedback resistor (R_f, 1–10 MΩ) sets the gain; a small feedback capacitor (C_f, ~2.2 pF) is required for loop stability given the photodiode's junction capacitance, calculated from the standard TIA stability relation: $C_f \geq \sqrt{C_d / (2\pi \times R_f \times GBW)}$, where GBW is the op-amp's gain-bandwidth product.

Two op-amp options are under evaluation for this stage, reflecting an explicit accuracy-vs-cost trade-off:

Option	GBW	Cost (est.)	Trade-off
OPA380 (precision TIA-optimized part)	~90 MHz	~Rs. 661/unit	Wide bandwidth preserves fast pulse shape faithfully; higher cost.
TL084 (general-purpose JFET-input quad op-amp)	~3 MHz	~Rs. 69/unit (4 amplifiers)	~16x lower bandwidth may attenuate/smear the fastest transient pulses; far lower cost and covers both detection channels from one chip.

The lower-cost option is being evaluated on the bench against measured particle-transit pulse widths before final selection, rather than assumed suitable by default — bandwidth requirements are directly dependent on flow velocity and beam waist, both of which will be measured empirically once the prototype is assembled (see Section 7).

5.4 Analog peak-detector

A fast switching diode (1N4148) and a small hold capacitor (C_{hold} = 10 nF) form a passive peak-detector: the diode charges the capacitor to the TIA output's peak voltage and blocks it from discharging as the signal falls. An N-channel MOSFET (2N7000), gated by a microcontroller GPIO through a pull-down-protected gate, discharges the capacitor after each ADC read, resetting the circuit for the next particle event.

5.5 Digitization and control

An ADS1115 16-bit ADC reads the held peak voltage over I2C (address 0x48), with a hardware ALERT/RDY interrupt line (pulled up to the logic rail) signaling conversion completion to the ESP32, avoiding inefficient polling. The ESP32 sequences laser triggering, peak-detector reset timing, and ADC reads, and is intended to run the acquisition loop as a dedicated task on one CPU core (with Wi-Fi/data reporting on the other) to avoid sampling being interrupted by network activity.

6. Bill of Materials and Cost Analysis

Component	Function	Cost (Rs., est.)
ESP32 DevKit board	Microcontroller, Wi-Fi, control logic	284
ADS1115 module	16-bit ADC, I2C interface	180
TL084 / OPA380 op-amp	Transimpedance amplifier	69 – 661
BPW34 photodiode	Optical detector	25
405 nm laser pointer pen (repurposed)	Excitation source + driver	280
2N2222 transistor	Laser switching	5
2N7000 MOSFET	Peak-detector reset switch	15
1N4148 diode	Peak-detector element	2
Resistors (assorted, ×10)	Feedback, pull-up/down, series	~20
Capacitors (assorted, ×5)	Feedback, hold, decoupling	~20
435 nm long-pass optical filter	Fluorescence isolation	Phase 2 item
Enclosure, tubing, misc.	Sample chamber, light-tight housing	Phase 2 item
Estimated total (core electronics, Phase 1, with safety margin)		~3,000 (est.)

Component substitutions (e.g., TL084 in place of OPA380) were selected through explicit cost-vs-performance analysis rather than by default, and are documented as a deliberate part of this project's engineering process, to be validated or reversed based on bench test results.

All costs in this document are estimates based on current hobbyist-electronics market pricing and are subject to change. The Rs. 3,000 total includes a deliberate safety margin over the itemized component subtotal, to account for shipping charges, potential reordering of parts damaged during hand-assembly (e.g., a fried op-amp), and price fluctuation across suppliers — rather than assuming a single best-case quote holds through the full build.

7. Planned Validation Methodology

The following test protocol is planned to characterize and validate the prototype, in order, before any biological classification claims are made:

- **Noise floor characterization:** record TIA output with the laser off, in a fully dark, sealed enclosure, to establish the baseline noise level against which real signals must be distinguished.
- **Op-amp bandwidth verification:** using a bench signal generator or a known reference pulse, confirm the selected op-amp's response time against actual particle-transit pulse widths at the system's real flow rate, before finalizing the TIA component choice.
- **Gain/linearity calibration:** characterize sensor output against particles of known, calibrated size (e.g., polystyrene microspheres) to establish a size-response curve and detection-limit threshold.
- **Biological vs. mineral discrimination test:** compare optical response of a known biological standard (food-grade baker's yeast, Saccharomyces cerevisiae — non-pathogenic, BSL-1) against a known mineral standard (e.g., fine silica or talc dust), across repeated trials, to quantify classification accuracy.

- **Repeatability testing:** repeated trials across multiple sessions, with statistical treatment (mean ± standard deviation across trials) to establish measurement reliability.

> *All biological validation work is restricted to BSL-1, non-pathogenic organisms (food-grade baker's yeast) and food-safe reagents. No pathogenic or antibiotic-resistant organisms are used in the current validation plan — see Section 10.*

8. Firmware Architecture

The ESP32 firmware is designed around three principles to ensure reliable, high-rate acquisition without data loss:

- **Dedicated acquisition task on one CPU core** (via FreeRTOS, pinned with xTaskCreatePinnedToCore), separate from Wi-Fi/MQTT networking on the other core, so network activity cannot stall sample timing.
- **Interrupt-driven ADC reads** using the ADS1115's hardware ALERT/RDY pin, rather than fixed-delay polling, for more deterministic timing.
- **Decoupled acquisition and transmission** via a FreeRTOS queue: the acquisition task pushes readings to a queue; a separate task drains the queue and publishes over MQTT, so a slow or blocked network call never stalls data capture.

9. Known Limitations and Open Engineering Questions

In the interest of technical honesty, the following limitations are explicitly acknowledged as part of this proposal, not discovered after the fact:

- **Weak native fluorescence:** unstained biological particles (e.g., unstained yeast) have relatively weak autofluorescence under 405 nm excitation at only 5 mW. It is possible that scattered excitation light leaking through the long-pass filter's roll-off edge may dominate the signal, in which case a cleaner separation may require a steeper/narrower filter, higher TIA gain, or a fluorescent viability stain — this will be determined empirically, not assumed.
- **Component-grade uncertainty:** the repurposed laser diode's exact electrical and optical specifications are not independently documented (salvaged from a consumer product), and its real output power will be measured post-extraction rather than assumed from the product's stated rating.
- **Species-level identification is out of scope:** the system distinguishes biological from non-biological particles based on aggregate scatter/fluorescence signature; it does not identify specific species, strains, or organisms.

10. Safety, Biosafety, and Compliance

- **Laser safety:** output power verified <5 mW (Class 3R) before enclosed use; wavelength-rated protective eyewear used during any manual alignment or bench testing with the enclosure open; fully enclosed, light-tight sample chamber during operation.
- **Biosafety:** all biological validation restricted exclusively to BSL-1, non-pathogenic organisms (food-grade baker's yeast) and food-safe reagents. No pathogenic organisms and no antibiotic-resistant strains are used or planned for use without explicit institutional biosafety approval obtained in advance.
- **Competition compliance:** any experimental work involving microorganisms will follow applicable science-fair Scientific Review Committee / Institutional Review Board approval processes prior to data collection, not after.

11. Future Research Direction (Explicitly Not Part of Current Prototype)

Everything in this section is a proposed, longer-term research direction and is not a claimed or demonstrated capability of the current or planned prototype. It is included to communicate the platform's broader potential, clearly separated from the validated, buildable scope described in Sections 1–10.

The same dual-channel (scatter + fluorescence) optical architecture is, in principle, extensible toward rapid phenotypic antibiotic-susceptibility screening — an approach aligned with published research directions in low-cost rapid-AST instrumentation. The underlying idea: rather than a single optical snapshot, a bacterial population is sampled at multiple time points (e.g., t = 0, 30, 60 minutes) after antibiotic exposure. Susceptible populations show measurable shifts in scatter/fluorescence distribution (indicating membrane damage or lysis); resistant populations remain optically stable relative to the untreated baseline. This is a population-level, time-course comparison, not a per-cell tracking method, since a flow-through optical system measures each particle once as it transits the beam.

This direction would require: (a) institutional biosafety supervision and approval before any live bacterial culture work, (b) validated non-pathogenic surrogate organisms for initial method development, and (c) substantially more validation data than a first prototype can generate, before any resistant-vs-susceptible discrimination claim could be responsibly made. It is presented here as the project's long-term motivation and growth path, not as current or near-term functionality.

12. Conclusion

OptiCyte's near-term goal — an approximately Rs. 3,000 (estimated) optical sensor capable of distinguishing biological from non-biological airborne particulates — is a scoped, buildable engineering project with a clear validation plan and explicitly acknowledged limitations. Its significance lies both in its immediate agricultural application and in its architecture as a credible, low-cost starting point for a class of instruments (optical flow cytometry) that is currently inaccessible outside well-funded laboratories — with a longer-term research trajectory toward rapid, low-cost antimicrobial susceptibility screening, pursued with the institutional rigor that work would require.

Prepared for ideathon evaluation. Costs are estimated in INR based on hobbyist-electronics market pricing at time of writing and are subject to change with sourcing and market conditions.